
ENERGY MONITORING AND CONTROL AUTOMATION SYSTEM USING IOT

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ABSTRACT

Automation can easily increase the efficiency of the Industries the main aim of our concept is to monitor and control the Industrial substation by using IOT. In earlier stage industries, power grids were monitored manually and now instead of that, here we using Internet. Main functions of IOT based automation is accurate fault location which avoids and eliminates error location. From anywhere in the world We can access information securely by using internet of things. IOT based automation system increase performance of electrical systems and immediate action can be taken to fix the problem once identified

Keywords: IOT, Automation, Monitor, Detector, Corrector

I. INTRODUCTION

In these days the electricity and power supply problems are impacting the performance of the industries. which can be avoided by implementing automation systems in industrial stations. . Main functions of IOT based automation is accurate fault location which avoids and eliminates error location. From anywhere in the world We can access information securely by using internet of things. IOT based automation system increase performance of electrical systems and immediate action can be taken to fix the problem once identified

The operation of electrical system can affected by changing the stability and load. To fix that by implementing automation systems for reducing that kind of error

The proposed system intends to assist an automation system for distribution feeders and the ring bus connected to them will be implemented and demonstrated. The proposed system covers various aspects both these sections of a grid like optimum utilization, safety, protection and switching to ensure best delivery of available power to the industries. A prototype of a two feeders connected to a ring bus will be implemented and the automation of the same will be demonstrated using an IOT systems and wireless communication, between the IOT system and the grid system. Using this system the protection from faults can be implemented. Also automatic identification of the faulty circuit can be implemented. Automatic intimation to ground staff can also be sent, instantaneously so immediate action can be taken to fix the problem once identified.

II. EXISTING SYSTEM

We propose and implement a system that has an automation system for distribution feeders and ring bus connected to it in order to reduce the failure of continuity of supply

BLOCK DIAGRAM

While doing a survey we found that the existing system is Electrical machinery with one – way communication and the generation is the centralized power generation and there is need of sensor to detect the failures and its almost manual monitoring and recovery. And also there are failures and power outages. And when it comes for users only few users can utilize the existing system.

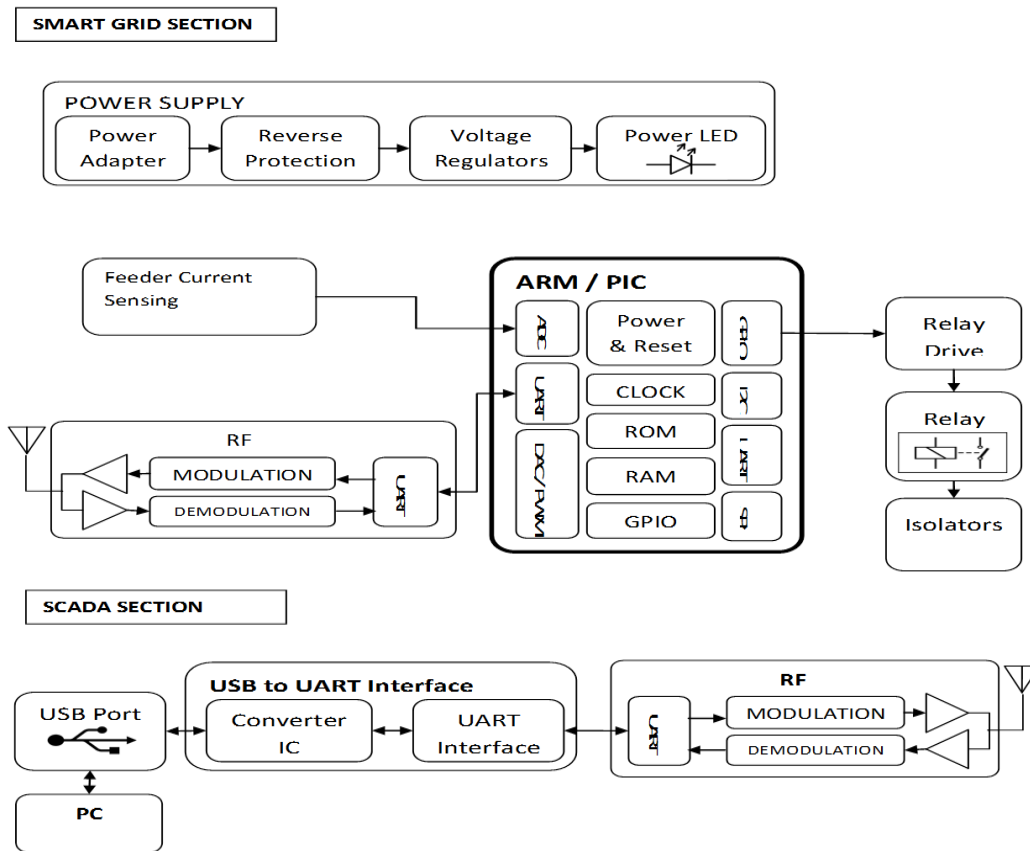


Figure 1: Existing system

DISADVANTAGES OF EXISTING SYSTEM

- Completely Manual operation
- Requires human attention and intervention at every stage and prone to human errors.
- Time consuming.
- Controlling and monitoring is done in a single place.

III. PROPOSED SYSTEM

BLOCK DIAGRAM

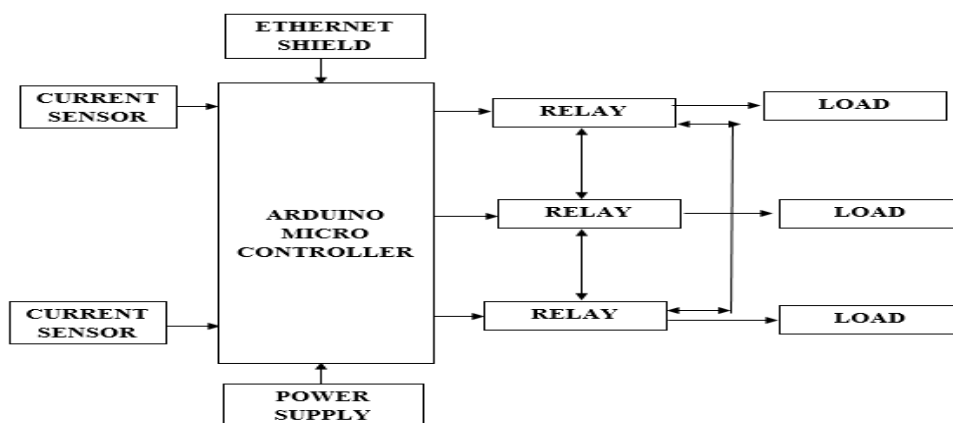


Figure 2: proposed system

EXPLANATION

The grid section consists of Feeder current sensing unit , Signal conditioning unit , Microcontroller and Relay. Feeder current sensing unit will sense the amount of input current flowing through the feeder. Current transformer is used for sensing current taken by the load. It is connected in series with the load. The interfacing circuit converts ac current into ac voltage .Therefore then the AC voltage is converted into dc voltage using bridge rectifier and filter. The DC voltage is then given to ADC. Then the value is send to the signal conditioning unit which manipulates an analog signal. Most common use is in analog-to-digital converters.

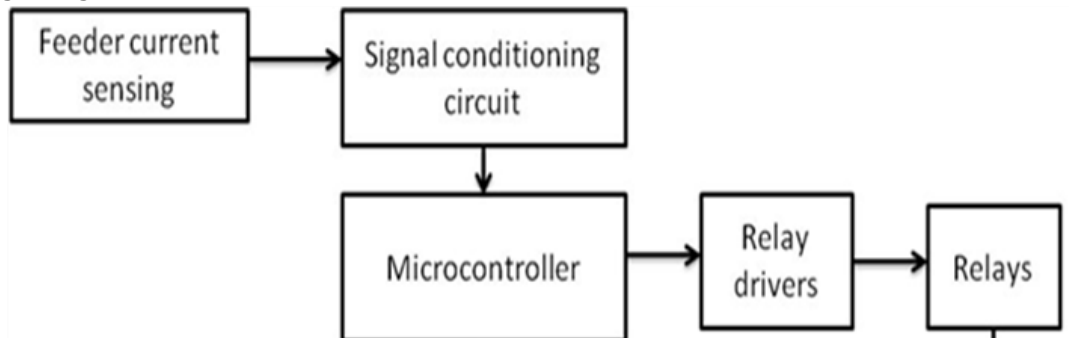


Figure 3 : Block Diagram of Grid Section.

Operational amplifiers (op-amps) are commonly used to carry out the amplification of the signal. Then the signal is given to the ARDUINO microcontroller which is used in this project. It is used for getting information from sensors and process it for automation of ring bus feeder and isolator is used for fault identification and isolating the faulted part from the circuit. After the fault is cleared or identified the information are sent to the IOT Section.

MODE OF OPERATION

There are two mode of operations,

- Under Normal Condition
- Under Fault Condition

Under normal condition

During Normal Condition the load is fed by the feeder 1 and relay for the feeder 1 is closed.

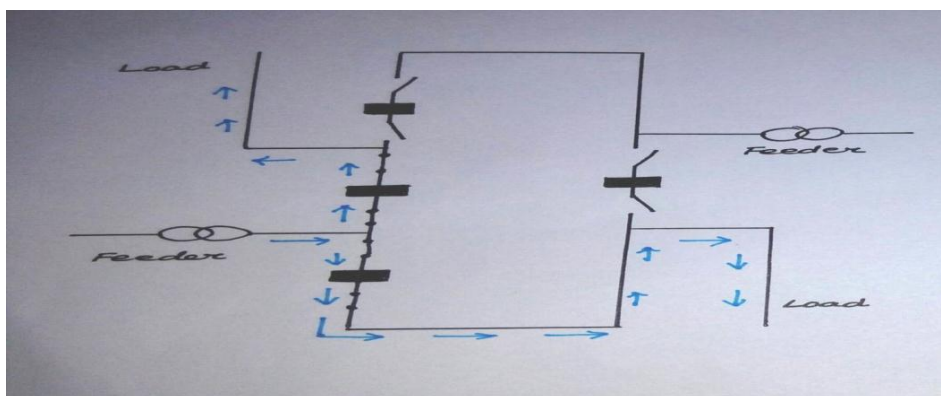


Figure 4 : Normal Condition

Under Fault Condition

Under fault condition, if there is any fault at the feeder the load is fed by feeder 2 and the relay 1 is opened therefore the feeder 1 is also opened. The fault may be due to environment conditions, current fluctuations etc.

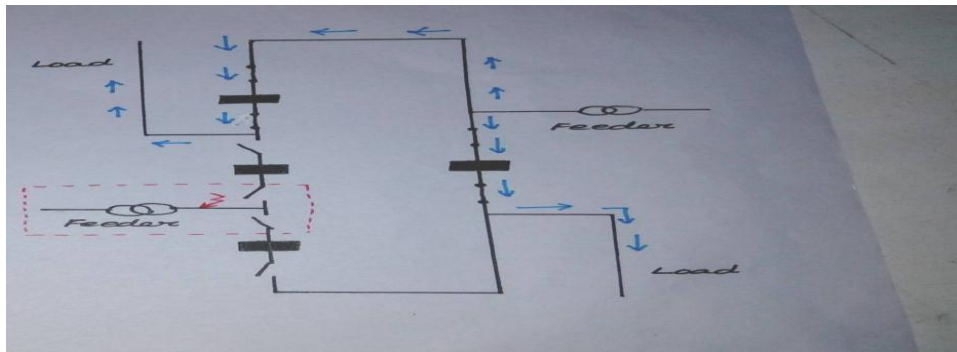


Figure 5: Fault Condition

Output table

Table 1: Output Table

FEEDER 1	FEEDER 2	FEEDER 3	FEEDER 1 LOADS		FEEDER 2 LOADS		FEEDER 3 LOADS	
			NORMAL	CRITICAL	NORMAL	CRITICAL	NORMAL	CRITICAL
ON	ON	ON	ON	ON	ON	ON	ON	ON
OFF	ON	ON	OFF	ON	OFF	ON	ON	ON
ON	OFF	ON	ON	ON	OFF	ON	OFF	ON
ON	ON	OFF	OFF	ON	ON	ON	OFF	ON

WORKING PRINCIPLE:

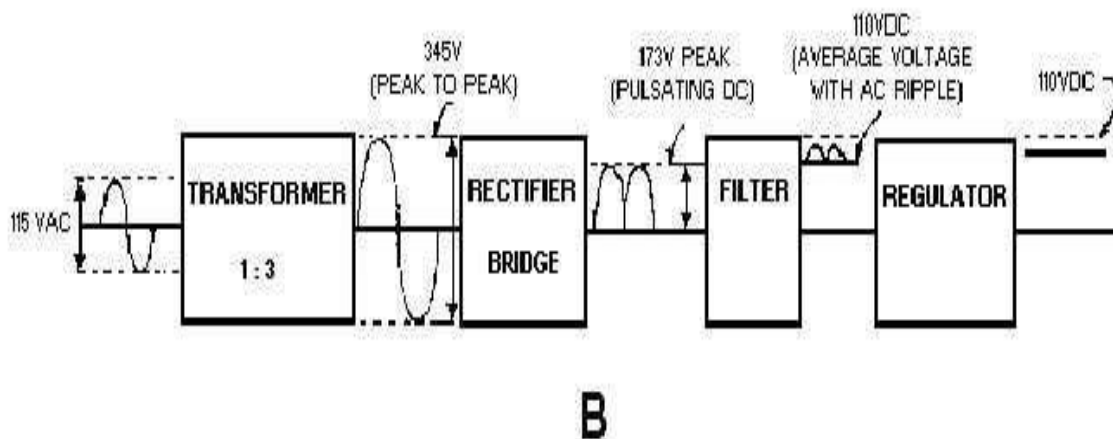


Figure 6: Block Diagram of Working Principle

The first section is the TRANSFORMER. It have steps up or steps down the input voltage and follows the EMI principle. The RECTIFIER section converts the AC signal to direct current. FILTER is used To block. The final section, the REGULATOR, it is used to maintains the output power at a stable level

IV. SIMULATION RESULTS

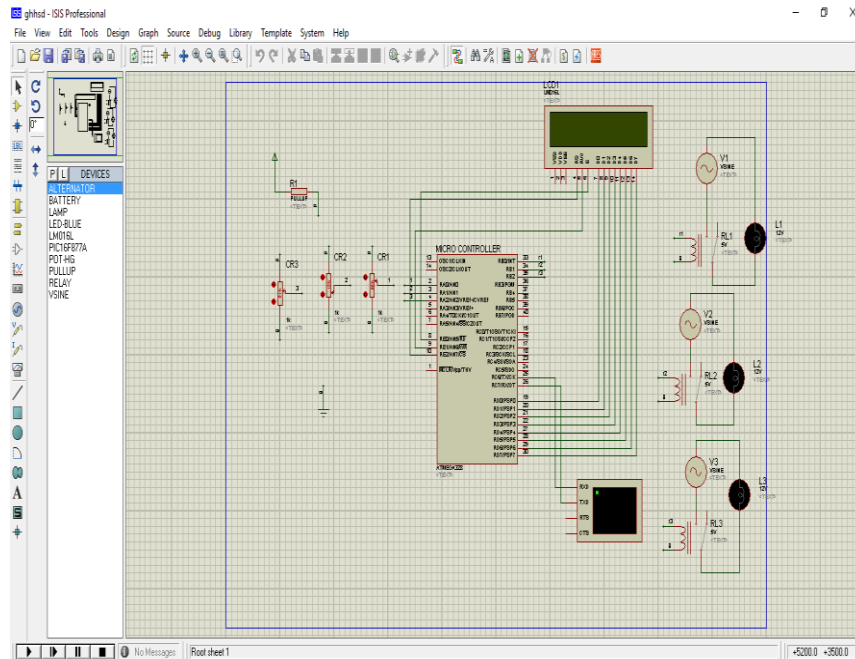


Figure 7: Simulation block diagram

As you can see in the above picture, the devices like relay, microcontroller, power supply, bulb, current sensor and lcd monitor have been picked from the proteus software library and their connections have been given according to our requirements and our required circuit is created.

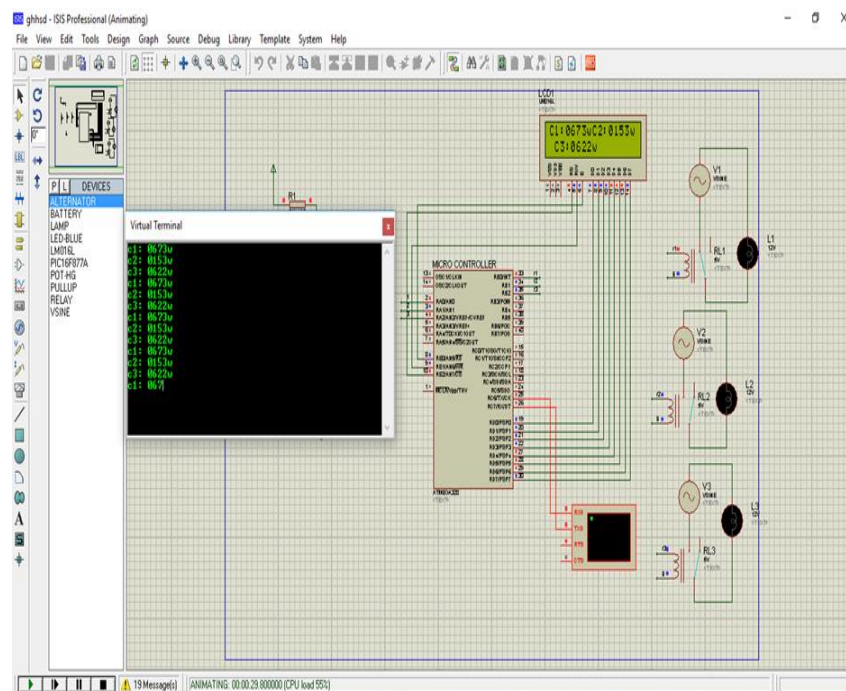


Figure 8: Relay 1 is ON

Now the simulation tool is made to run by clicking the play button in the bottom of the screen and you can see the virtual terminal gets opened and the LCD display starts to show the current values.

The Relay 1 is turned ON by giving command to the microprocessor by Pressing the key on the number 1 and the same will be followed to the balance relays.

V. APPLICATION AND FUTURE DEVELOPMENT

Application

These kind of smart technologies are used in other departments such as telecommunication and several operations such as

- Integrated communication.
- Sensing and Measurement.
- Smart meters Phasor measurement units.
- Power system automation

Future Development

- More feeders can be utilized for smart power distribution.
- And many other parameters other than voltage and current can be detected.
- Self-Healing capability can be improved.
- Efficiency of the automation technology therefore can be improved.

VI. CONCLUSION

Thus we have developed IOT based distribution feeder system for automated distribution of power according to the power required by the load and also the fault isolating in the circuit. The ring bus system improves the quality and reliability of power supply and also reduces the power losses. The fault detection, isolation and restoration process is very fast in the grid distribution system. Overall, the approach is found to be accurate, simple to implement and results in quick reaction to sudden changes of load especially when there is power fluctuations. Thus by making the grids smarter and automated the power loss could be reduced, the components can be prevented from damage and all other components can receive power except the component causing overload.

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